Answer Key

Grade 6 Mathematics · Unit 9: Two-variable Relationships · Answers for every item on Sheets 1 and 2 of each lesson packet. Constructed responses show a model answer; accept equivalent reasoning.

LESSON 9-1

Independent and Dependent Variables · Tables of Values

Sheet 1 · Independent and Dependent Variables

1. a. minutes biking I, miles traveled D b. total cost D, tickets bought I c. guests I, cups of juice D

2. Model: "Her distance from home depends on the time she rides the bus." (time riding = independent, distance from home = dependent)

3. Independent: number of dogs walked. Dependent: money earned (it depends on the number of dogs).

4. a. weeks (on the x-axis) b. height in inches (on the y-axis) c. After 4 weeks, the sunflower is 20 inches tall.

5. The size of a number does not decide the variable. The sandwiches needed depend on the number of people, so the number of people is independent and the number of sandwiches is dependent.

Sheet 2 · Tables of Values

1. 8, 10, 12, 15

2. $8, $11, $14, $23

3. Multiply the boxes by 8 (y = 8x); 4 boxes: 32 markers, 8 boxes: 64 markers

4. The last value, 8, is wrong. When x = 4, y = 2 × 4 + 1 = 9 (the y values go up by 2 each time).

5. $6, $10, $14; $20 is 8 miles (2 × 8 + 4 = 20)

TEST PRACTICE

6. C – 29

7. B

LESSON 9-2

From Table to Graph · Analyzing a Graph

Sheet 1 · From Table to Graph

1. a. (4, 12) b. (0, 0) c. (6, 30) d. (9, 2)

2. $4, $8, $12, $20; points (1, 4), (2, 8), (3, 12), (5, 20) on a straight line through the origin

3. a. (5, 10) b. After 3 hours, Kofi has hiked 6 miles. c. 2 miles per hour

4. (1, 6), (3, 18), (4, 24). (0, 0) means 0 hours of renting costs $0.

5. He switched x and y. The x value (hours) comes first: plot (2, 10) and (4, 20).

Sheet 2 · Analyzing a Graph

1. a. After 4 minutes, the pool has 60 gallons. b. 15 gallons per minute c. 135 gallons

2. a. 7 b. 8 c. 12

3. Unit rate: 18 ÷ 3 = 6 pages per hour; 7 hours: 6 × 7 = 42 pages

4. Job A: 24 ÷ 4 = $6 per hour. Job B: 18 ÷ 6 = $3 per hour. Line A is steeper, so Job A pays more each hour.

5. No. A proportional graph must pass through (0, 0); this one starts at (0, 10). (0, 10) means she had $10 saved at the start.

TEST PRACTICE

6. C – 7

7. B

LESSON 9-3

Equations from Tables · Equations from Situations

Sheet 1 · Equations from Tables

1. a. y = 7x b. y = x + 9 c. y = 5x d. y = x − 3 e. y = x + 6 f. y = 9x g. y = 8x h. y = x − 5

2. y = 11x; 66, 99

3. Each y is 2 less than x: y = x − 2. When x = 20, y = 18.

4. y = 4x; 10 tickets cost $40

5. It only works for the first row: 3 × 4 = 12, not 8. Each y is 4 more than x, so y = x + 4.

Sheet 2 · Equations from Situations

1. a. y = 12x b. y = x + 3 c. y = 15x + 20

2. y = 5x + 4; 4 games: 5 × 4 + 4 = $24

3. y = 2x + 3; 7 miles: 2 × 7 + 3 = $17

4. $30, $38, $46, $70; y = 8x + 30

5. She added the fee to the rate, so every hour costs $35. The fee is paid once: y = 10x + 25.

TEST PRACTICE

6. B

7. C

LESSON 9-4

Substitute and Solve · Working Backward to a Goal

Sheet 1 · Substitute and Solve

1. a. 19 b. 72 c. 28 d. 42

2. 3, 5, 7, 11, 13; points (0, 3), (1, 5), (2, 7), (4, 11), (5, 13) on a straight line

3. a. 15 × 6 + 20 = 110 gal b. 15 × 10 + 20 = 170 gal

4. 3 hours: 7 × 3 + 5 = $26; 8 hours: 7 × 8 + 5 = $61

5. a. Yes: 5 × 4 + 2 = 22 b. No: 5 × 6 + 2 = 32, not 30

6. He added before multiplying. Multiply first: 4 × 5 = 20, then 20 + 3 = 23. y = 23.

Sheet 2 · Working Backward to a Goal

1. a. 9 b. 13 c. 9 d. 8

2. 210 − 45 = 165; 165 ÷ 15 = 11 weeks

3. 150 − 30 = 120; 120 ÷ 12 = 10 weeks. Check: 12 × 10 + 30 = 150.

4. a. week 4 b. 33 − 6 = 27; 27 ÷ 3 = 9, so week 9

5. 150 ÷ 12 = 12.5. She cannot buy half a box, and 12 boxes (144) is not enough, so she buys 13 boxes.

TEST PRACTICE

6. B – 6 hours

7. C – x = 13